Closed meander counts M35-M38

Factual evidence prepared for OEIS A005315 | Efthimios Alepis | ORCID 0000-0002-3862-7218

This report accompanies a candidate replacement b-file. The public OEIS entry was checked through n=34 when assembled; the four additional values remain subject to OEIS editorial review.

Exact values

M_{35} = 211433268000094602372146238438566

M_{36} = 23582618387517290192123441049260 12

M_{37} = 263706529694495108792090969132427 92

M_{38} = 29559808182280424672448776069099 4952

Order	Measured wall time	Peak sampled RSS	Matrix I/O	Numerical check
M35	31,603.365 s	828,465,958,912 B	0 / 0 B	2 modular passes
M36	2,191.354 s	945,597,001,728 B	0 / 0 B	2 modular passes
M37	5,689.257 s	1,771,293,687,808 B	0 / 0 B	11 family passes
M38	12,387.379 s	1,599,349,862,400 B	0 / 0 B	No CRT

The runs used 16 numerical OpenMP threads, but different code revisions and orders. The listed times are measured wall times, not a controlled speed comparison. Matrix I/O excludes ordinary logs and compiler cache traffic. RSS is physical process memory, not the coefficient-array allocation estimate.

Method

A closed meander of order n is counted as a pair of noncrossing perfect matchings on 2n ordered crossings whose union is one cycle. The program constructs weighted lower-family tree states, sparse upper/lower frontier geometry and cropped bitmap supports. Exact coefficient transfers advance the frontier, and a finite eight-crossing terminal operator closes the count. M38 divides the lower families into four groups and uses rank-64 geometry.

Priority and relation to earlier work

Vladimir Osipov reported M29 and M30 earlier; this work makes no priority claim for those terms. The earlier C5/V2 source is a separate lineage. The tree/bitmap repository is intended to provide executable source, unchanged executed source manifests and raw numerical records for the new terms.

Verification details

M35-M36

Each exact count was reproduced by two modular runs, modulo 2^64 and 2^64-1. Their coprime product exceeds the relevant Catalan-pair upper bound, so the combined residues uniquely reconstruct the nonnegative count in that range. The modular runs reuse tree grammar and frontier geometry; they are numerical checks, not a structurally independent algorithm.

M37: 11 modular family passes

Four disjoint lower-family groups were evaluated in fresh modular runs. Groups 1-3 used rings modulo 2^64, 2^64-1 and 7; group 4 used the first two. Each run recorded 66 stages and zero matrix I/O. Each group's CRT product exceeds its stated family upper bound. The reconstructed group values sum exactly to M37.

Group	Reconstructed contribution	Passes	Largest pass RSS
G1	567101144859362468731568792622343 6	3	886,711,066,624 B
G2	862719759107092287358678567725144 4	3	1,031,501,447,168 B
G3	816770689885259780818450817875337 2	3	1,250,442,362,880 B
G4	390473703093236551012211513101454 0	2	842,644,873,216 B

Peak sampled RSS across all 11 modular passes: 1,250,442,362,880 B. These passes share the tree grammar and bitmap geometry with the exact M37 program. They therefore do not independently prove those shared components.

M38: four exact groups

Group	Exact contribution	Wall time	Peak RSS
1	530221880564408645809468488955515 42	2825.560 s	1,254,581,493,760 B
2	773718476008210905359425367711240 28	2708.577 s	1,363,339,526,144 B
3	128106091870808593177303427240780026	3147.498 s	1,551,855,460,352 B
4	370979542947336984302949477835393 56	3700.194 s	1,599,349,862,400 B

The 4 contributions sum to 29559808182280424672448776069099 4952. All 272 stages completed. The measured controller session took 12,387.379 s, and the maximum sampled RSS was 1,599,349,862,400 B. No independent modular CRT has yet been performed for M38.

Evidence and limits

The public source release contains the frozen executed source packages, their SHA-256 manifests, per-stage raw records, exact and CRT result JSON, an integrity checker, and the candidate OEIS b-file. SHA-256 confirms distributed byte integrity; it is not institutional certification or proof that a remote run occurred. The M38 result is supported by one completed exact four-family run and internal tests. No formal proof of every software path or new asymptotic complexity bound is claimed.

References

OEIS A005315: https://oeis.org/A005315 . Tree/bitmap source: https://github.com/talepis/meander-tree-bitmap (verify public availability). Earlier C5/V2: https://doi.org/10.5281/zenodo.23687622 . Vladimir Osipov's earlier M29/M30 report: https://doi.org/10.5281/zenodo.22773575 .